Lab experiment 1

Aim

To write a Python program using Regular Expressions to match and search for patterns in text.

Algorithm

1. Start the program.
2. Import the `re` module.
3. Define a text string containing words and numbers.
4. Use `re.search()` to search for a pattern in the text.
5. Use `re.match()` to check the pattern at the beginning of the text.
6. Use `re.findall()` to find all occurrences of a pattern.
7. Use `re.I` for case-insensitive searching.
8. Display the results.
9. Stop the program.

Code :

```
import re

text = "Python is easy. I am learning Python and Java."

# 1. Search for a pattern

result = re.search("Python", text)

print("Search:", result.group() if result else "Not found")

# 2. Match at the beginning

result = re.match("Python", text)

print("Match:", result.group() if result else "Not found")

# 3. Find all occurrences

result = re.findall("Python", text)

print("Find All:", result)
```

```
# 4. Case-insensitive search

result = re.findall("python", text, re.I)

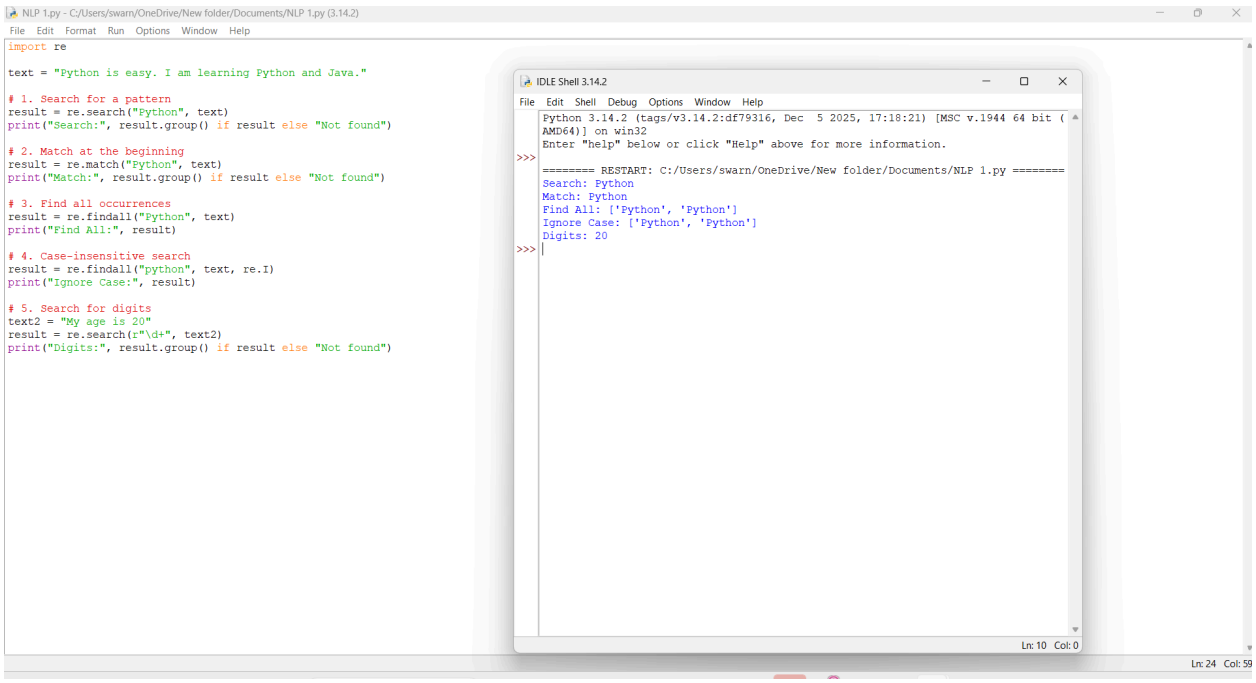
print("Ignore Case:", result)

# 5. Search for digits

text2 = "My age is 20"

result = re.search(r"\d+", text2)

print("Digits:", result.group() if result else "Not found")
```



Result

Thus, the Python program to demonstrate Regular Expressions using `re.search()`, `re.match()`, and `re.findall()` was executed successfully, and the required patterns were matched and searched in the text.